

An Explainable Hybrid Machine Learning Framework for Loan Approval Prediction Based on Random Forest, Light Gradient Boosting Machine, Combined with Grey Wolf Optimizer Algorithms

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Abstract

Loan approval prediction remains one of the most important problems in financial decision-making within financial institutions. In recent years, the increasing volume of credit applications has made loan approval prediction a central topic in both research and practice in credit risk management. This paper presents an explainable hybrid machine learning approach for predicting whether a loan application will be approved or rejected. The entire machine learning model development process is structured based on the well-established CRISP-DM process model. In this study, a publicly available loan approval prediction dataset obtained from Kaggle was used. The raw dataset was first subjected to outlier analysis, transformation of categorical variables into numerical representations, and Min-Max normalization. In the experimental evaluation, nine different classification algorithms, namely Decision Trees, Random Forest, Gradient Boosting Machines, XGBoost, LightGBM, CatBoost, Support Vector Machines, Naive Bayes, and K-Nearest Neighbors, were comparatively evaluated in terms of predictive performance. To improve the predictive performance of the candidate models, their hyperparameters were optimized using the Grey Wolf Optimizer (GWO), and model-selection performance was evaluated through 3-fold nested cross-validation. The obtained results show that state-of-the-art ensemble-based algorithms clearly outperformed more conventional classifiers. During the model-selection stage, the GWO-optimized LightGBM model achieved the highest 3-fold nested cross-validation accuracy among the candidate models, with an accuracy of 91.13%. Following this model-selection procedure, the final RF-LightGBM-GWO hybrid model achieved an accuracy of 92.17% on the independent/unseen 20% hold-out test set. Since most ensemble learning models are inherently black-box models, SHAP-based Explainable Artificial Intelligence (XAI) was employed to provide insight into the decision-making process of the models. The results demonstrate that advanced ensemble learning techniques, metaheuristic-based hyperparameter optimization, and explainable AI can be integrated into a single coherent analytical pipeline.

Keywords: *Grey Wolf Optimizer, hyperparameter optimization, LightGBM, loan prediction, Random Forest Algorithm, XAI*

1. Introduction

Credit scoring is an important tool for lenders to determine whether or not to grant credit to a borrower and what terms to offer. In a score, lenders obtain a simple reference, whether numerical or alphabetical, that describes the score of a borrower as positive or negative. As the consumer and commercial credit markets are rapidly growing, established models such as scorecards and regression models are still being heavily utilized for underwriting. These models provide accurate, consistent, reliable, and efficient results which are easy for lenders to understand and act upon. Yet, the increasingly diverse, high-dimensional, and non-linear nature of lending data poses challenges to scorecards. Loan underwriting and subsequent credit scoring in the classical approach involved a lot of manual assessment, a lot of expert judgment and simple statistical models. Fundamentally, this amounts to regression and scorecards, yet their limited potential to handle the complex, non-linear, and diverse data that modern credit data represents has gradually given way to the adoption of machine learning (ML) techniques, and their performance has been benchmarked and reviewed. Importantly, ensemble methods have been shown to significantly outperform single models. As the global economy precariously perches on the edge of recession, risk and portfolio managers are under pressure as never before to measure and manage risk. Advanced analytics are becoming essential to independent risk assessment and to optimal allocation of capital [1–4]. Existing systems for customer credit assessment have several inherent deficiencies. Within this context, the

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potential of ML methods for improving the assessment of a customer's creditworthiness is investigated, and an optimized framework for fully exploiting this potential is proposed and tested in order to make more informed grant or refusal of credit decisions.

Our goal is to create a full pipeline for data mining for the task of predicting credit approval, and to employ several ML methods in a unified framework of hyperparameter optimization. Importantly, satisfactory accuracy is not the only thing we want to achieve - we also want to know how the underlying artificial intelligence came to a given decision. To get the most out of current algorithms, we built this data mining pipeline to extend their capabilities to better tackle the problem of determining credit risk. Data from actual loan approvals (Section 4) is used to test the pipeline. We examined the best-performing models to determine whether loan decisions should be made. This required us to analyze various borrower characteristics. Subsequently, we integrated these models by thoroughly analyzing the features that explain the decision-making process.

When the framework is applied in practice, a clear potential for improving decision-making accuracy in credit approvals can be observed. In this way, financial institutions that aim to refine their risk management strategies are provided with a practical and reasonably transparent tool, instead of being left with yet another black-box solution. Moreover, ML performance evaluation metrics are highly vulnerable to hyperparameter tuning. Thus, making hyperparameter optimization a must component of effective model development [5]. In response to these methodological considerations, the present study develops a comparative framework for loan approval prediction through evaluating multiple machine-learning algorithms—Decision Tree, Random Forest, Gradient Boosting Machine, XGBoost, LightGBM, CatBoost, Support Vector Machines, Naive Bayes, and K-Nearest Neighbors under a combined hyperparameter optimization and feature selection strategy based on the Grey Wolf Optimizer (GWO). In this study, we seek to identify the most appropriate models for predicting the credit approval classification task by leveraging multi-model benchmarking, metaheuristic parameter tuning, and feature selection.

2. Literature Review

Credit application approval is a fundamental process within financial institutions, essential for effective risk management and ensuring the sustainability of their credit portfolios, given that the primary source of credit risk stems from the potential default of borrowers. One of the principal tools of analytical models used in lending for several decades is the credit score. Developed by specialty companies that provide scores calculated from the information contained in a borrower's credit file, the score purportedly measures the likelihood that a borrower will repay loans on time. Statistical classification techniques such as scorecards, logit, and linear probability models have been used to develop these scores and to distinguish between good borrowers and those who are less creditworthy. For several decades, scorecards and traditional regression models built using methods of classical statistics have dominated the field. These models are familiar, easy to understand, simple to integrate into an organization's underwriting process, and well-suited to the repetitive, high-volume environment in which most underwriting is conducted. In recent years, however, the use of alternative approaches for modeling creditworthiness has grown. The growing volume of available information, the increase in dimensionality of the data, and the complexity of the structures contained in the large pool of available data on the various aspects related to the credit (credit-related data) have caused a relevant shift in the literature from statistical methods to ML (Machine Learning) techniques, in order to capture nonlinear relationships and interactions between different variables. In the last few years, many benchmarking papers and reviews have emphasized the use of even more powerful classifiers in order to improve the performance of models for credit risk assessment. In recent years, ML methods have been deeply affecting the field of credit scoring, and in particular, the default prediction on loans. Being more accurate, more powerful, and data-driven than traditional statistical methods for credit risk modeling, ML techniques have started to rewrite the rules for credit risk modeling by adopting non-trivial, non-linear functional forms. In addition to time series forecasting, there is also a growing interest in applying ML (ML) models. This is because ML models are known to outperform traditional methods in complex and unstable domains like finance [2,3,6].

It should be mentioned that there is a comprehensive systematic review of papers and articles that have applied ML techniques to the task of credit scoring. The review includes papers from around 2018 to 2024. Boosted tree and random forests were the most common techniques used in these papers. However, many other ML techniques were also explored and, in many cases, outperform traditional methods like logistic regression in terms of accuracy and robustness. A total of 63 studies were included in the review. The author comments on current trends in the field and observes that a few of the same problems crop up no matter what particular approach is being utilized. The review provides a theoretical map for the field; this work uses that map rather than starting from scratch [7].

Furthermore, in empirical applications, another study examines the application of ML to credit risk management and demonstrates how large datasets and complex risk metrics can be used to reduce losses due to default and improve the accuracy of credit-granting and pricing decisions above and beyond that achieved by scorecards. [8].

While there is ongoing debate about the conceptual foundations of assessing and managing credit risk in the new credit market, several studies have successfully applied ML (ML) models in assessing and predicting credit risk and default probability. In one study, k-nearest neighbors (KNN), support vector machines (SVM), and random forests were employed in developing a framework for evaluating credit risk and default probability for online peer-to-peer lending, and it was found that ML models outperform traditional logistic regression in terms of accuracy and discrimination power as measured by AUC and Brier scores [9].

Besides the applications of ML methods in general, there have been numerous studies on predicting credit card default from a domain-specific perspective. In order to improve the accuracy of credit card default prediction, many methods and models, including support vector machines (SVM), decision trees, bagging, boosting, random forests, and gradient boosting, have been adopted. In addition, many studies applied the stacked generalization method to default prediction. From a domain-specific perspective, this paper reports experiments on the prediction of customers who would default on their credit card payments using the gradient boosting method, XGBoost, and other methods. The results from various data sets indicate that current ML methods are sufficiently effective for a variety of credit risk analyses, including credit card default prediction [10].

Another attempt to evaluate the performance of some of the most recent ML algorithms on loan data. The key differences to similar attempts are the usage of Neural Network and several of the most recent versions of boosted decision trees, XGBoost, Gradient Boosting, Random Forest, and LightGBM. Categorical variables are encoded, additional feature engineering is applied, and there is a technique used to deal with the class imbalance problem. The results of the evaluation on real data show a significant improvement in the performance of Gradient Boosting and XGBoost compared to traditional methods for the task of assessing credit risk and classifying loans as credible or not [11].

In the context of benchmarking traditional against ML approaches, one of the recently carried out studies compares the performance of traditional models and ML models on benchmark datasets. They employed deep learning neural network models and tree-based ensemble models and compared their performance with traditional models such as logistic regression and discriminant analysis on credit default datasets. The experimental results on multiple datasets show that the ML models outperform the traditional models on all metrics, including F1 score, G-mean, and AUC for a credit scoring task, i.e., whether a credit card holder will default [12].

Collectively, these studies illustrate that ML can significantly improve the accuracy and ability to handle and model complex relationships, including nonlinearity and adaptability, of modern credit risk assessment models. Using techniques such as ensemble learning, gradient boosting provides both theoretical justification and empirical evidence of the usefulness of ML for these important business applications.

Empirical evidence in the credit scoring literature consistently demonstrates that ML models—particularly tree-based methods such as random forests and gradient boosting—outperform traditional statistical approaches in terms of predictive accuracy, discrimination power (e.g., AUC), and overall robustness [3,6,13].

Machine-learning methods are particularly effective in modeling heterogeneous financial behaviors and risk profiles, as they can capture nonlinear relationships, accommodate high-dimensional data, and learn complex interactions among borrower characteristics, often with stronger predictive performance than traditional linear models [7,14].

It should be especially underlined that, while choosing a suitable ML algorithm is important, much more impactful benefits and gains can be achieved by improving how one preprocesses, engineers features, and evaluates models. Many top-performing models in recent literature include ensemble and hybrid approaches, and in this study, the relatively new CatBoost algorithm is explored for classification tasks [15].

At the current state of the art, research on credit scoring has become more focused on making ML models understandable and relatable to people, rather than just comparing their predictive power. In the recent research landscape, researchers value models that not only predict accurately but also can clearly explain how they arrive at their decisions. Explainability of Artificial Intelligence (XAI) has recently gained significant traction in Finance, particularly in credit evaluation. Methods and techniques such as SHAP and LIME have started to garner significant attention and have been employed in a variety of ML contexts in order to garner insight into how these models are making lending decisions. These tools help make the models more transparent, support regulatory oversight, and foster trust—without sacrificing the strong predictive capabilities of advanced ensemble techniques [16,17]. In particular, SHAP has emerged as one of the most widely used explanation tools in credit-risk research

because it can generate both global and local explanations, allowing researchers to identify the most influential variables. It is also interpreting individual approval or rejection outcomes [18,19].

More recent studies published or circulated in 2025–2026 publication venues, together with recent 2026 preprints, further indicate that credit-risk modeling research is moving beyond simple algorithm benchmarking toward validation reliability, data-leakage control, class-imbalance handling, multimodal information sources, and deployment-oriented robustness. Another important review provides [20] a recent systematic literature review of machine-learning-based financial credit scoring and emphasizes that ML models are increasingly used to improve borrower classification, while also highlighting persistent challenges related to interpretability, bias, dimensionality, and the standardization of datasets and evaluation metrics. Similarly, a recent study reviews [21] deep-learning approaches for credit risk prediction and indicates that recent research has expanded from classical tabular models toward architectures capable of capturing nonlinear, temporal, and relational dependencies in borrower-level credit-risk data. From a methodological perspective, [22] specifically highlight the importance of leakage-aware feature selection, strict temporal partitioning, and class-imbalance treatment in mortgage default prediction, reporting that an AutoML-based approach achieved the highest AUROC among the models compared in their experimental setting. In addition, [23] demonstrate that interpretable textual signals extracted from 10-K narratives can provide incremental information for bankruptcy-risk prediction beyond conventional accounting ratios, increasing the AUC from 0.8323 to 0.9019 in their one-year hold-out test. Complementing these prediction-oriented studies, [24] frame financial AI as a continuously operated lifecycle involving training and updating, deployment and inference, and operation with monitoring and feedback, thereby emphasizing that financial AI systems should be assessed not only in terms of predictive accuracy but also in terms of robustness, security, and downstream decision consequences. Taken together, these recent studies support the need for credit approval prediction frameworks that combine predictive performance with careful validation design, leakage-aware modeling, interpretability, and operational reliability.

Recent applied literature also reinforces the practical relevance of combining advanced machine learning, explainable AI, and business-oriented decision support in credit-risk-related applications. [25] proposed an LLM-FP-CatBoost framework for fintech lending, in which large language models are used to extract narrative information as supplementary credit data, while a FocalPoly loss is integrated with CatBoost to address class imbalance; importantly, SHAP is also used to improve the interpretability of the proposed model. [12] compared traditional statistical models with modern machine-learning methods for credit card default prediction and reported that modern approaches outperform traditional models in terms of predictive indicators such as F1-score, G-mean, and AUC, with the deep neural network achieving the strongest performance among the modern ML models examined. From a broader business perspective, [26] discusses the increasing use of AI applications in areas such as customer analytics, decision support systems, data analysis, anomaly detection, fraud detection, and risk management, which strengthens the managerial relevance of AI-based credit evaluation systems. In a more domain-specific business-risk context, [27] developed an AI-based customer risk classification framework for receivables management and reported that XGBoost achieved the highest classification accuracy in both the Scikit-learn simulation and the MindsDB real-time database implementation. Furthermore, it has been demonstrated that, in an explainable machine-learning setting outside the direct credit-risk domain, feature-selection-based modeling can preserve high predictive performance while improving interpretability by clarifying which selected features contribute to model decisions [28]. Although these latter studies are not all direct credit approval studies, they provide recent methodological and applied evidence that AI-based classification systems are increasingly expected to be accurate, explainable, computationally manageable, and relevant for real-world decision-support contexts.

Despite these advances, evidence remains limited for integrated credit-approval frameworks in which multiple contemporary classifiers are compared, optimized under a unified hyperparameter-search strategy, and interpreted through explainable AI within a single end-to-end pipeline. In many previous studies, either predictive performance has been prioritized without a common optimization design, or interpretability has been introduced only after model development for selected classifiers, rather than being embedded in a systematic multi-model framework [7,16]. In response to this gap, the present study evaluates multiple machine-learning algorithms using Grey Wolf Optimizer-based hyperparameter tuning and incorporates explainable AI into the final analysis, thereby offering a more transparent, comparable, and methodologically coherent framework for credit approval prediction [16,29].

3. Methodology

In this study, we have followed the Cross Industry Standard Process for Data Mining (CRISP-DM), a process model with six phases that describes the data science life cycle [30,31]. In Figure 1, all phases of the CRISP-DM are illustrated.

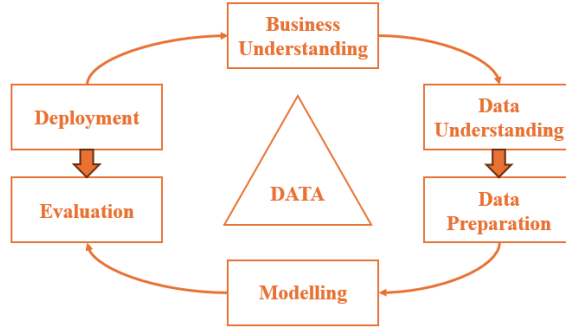


Figure 1: Phases of CRISP-DM (Shearer, 2000).

3.1 Overview of the Modeling Framework

In this study, a comparative ML framework is developed to evaluate the performance of multiple classification algorithms for credit approval prediction. The selected models will cover a range of ML techniques including Tree-Based Ensembles and more traditional models to test their ability to handle strong relationships, specifically non-linear relationships, as well as overall model performance and robustness. Let the dataset be defined as

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \quad (1)$$

where $x_i \in \mathbb{R}^p$ represents the feature vector of the i -th applicant and $y_i \in \{0,1\}$ denotes the credit approval outcome.

3.2. ML Models Used in the Study

3.2.1. Decision Trees

Decision Trees is a supervised learning algorithm that trains a tree-like model that splits the space of the input features in order to predict a target variable in both regression and classification tasks. For a classification problem, when building Decision Trees, the space of features is recursively partitioned to create regions that are more homogeneous than the input space in order to make accurate predictions on the target variable. In the splitting process of the subspaces of the feature space, a decision tree determines the quality of a potential split, this quality being often measured by an impurity function, the most commonly used one being the Gini index – particularly CART Decision Trees [32–34] or entropy-based information gain in ID3 and C4.5 algorithms [35–37]. In this context, the Gini impurity is defined as

$$G = 1 - \sum_{k=1}^K p_k^2 \quad (2)$$

where (p_i) denotes the proportion of observations belonging to class (i) in a given node. The recursive splitting process continues until a predefined stopping criterion is reached, such as the maximum tree depth or the minimum number of samples required in a leaf node. Decision Trees are highly interpretable and can effectively capture nonlinear relationships and feature interactions among predictors. However, when used as standalone classifiers, they are prone to overfitting, high variance, and instability, which may limit their generalization performance in complex prediction tasks [32,34,38].

3.2.2. Random Forest Algorithm (RF)

Random Forest (RF) is an ensemble learning method used to improve accuracy and robustness/generalization performance of classification and regression decision rules. Ensemble decision trees are generated; each tree is constructed on a bootstrap sample of the training data (a random subset of the available instances, possibly with

duplicates). During the building of each tree, a random subset of features is considered at every split point, which introduces variability among the trees and reduces correlation. The final prediction is obtained by aggregating the outputs of all individual trees, typically through majority voting for classification tasks or averaging for regression tasks, thereby achieving improved generalization over a single decision tree [39]. It is highly essential to emphasize that RF improves the predictive performance of single decision trees by combining multiple trees constructed on different bootstrap samples of the training data. When it comes to classification problems, the final prediction is obtained by majority voting across the ensemble:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), h_3(x), \dots, h_B(x)\}, \quad (3)$$

where $h_B(x)$ denotes the prediction of the B -th tree and B is the total number of trees. Random Forest is a common learning method that reduces both the training data and the features, and thus improves the generalization performance and prevents overfitting of single decision trees. Due to its robustness, the possibility to learn highly non-linear relationships and also the noise robustness, Random Forest is often used as a benchmark classifier for credit risk assessment problems and also for loan applications.

3.2.3. Gradient Boosting Machine

Gradient Boosting is a method for building a strong prediction model by forming a sequence of relatively weak models, added up “with boost” to create a much better model. This is commonly achieved by forming many decision trees, where each tree is relatively small (i.e., shallow). But in Gradient Boosting, each model is fit to the negative of the gradient of a differentiable loss function (e.g., squared error for regression, logistic loss for classification), evaluated at the current prediction. By repeatedly improving the predictions of the current models, Gradient Boosting achieves high accuracy for many types of prediction tasks (including, but not limited to, those with complex structure and those with tabular data (Friedman, 2001). Distinguished from many other techniques which directly attempt to fit the target variable for a new learner, in Gradient Boosting the new learner is fit to the negative gradient of the loss function with respect to the current model’s predictions, i.e. the pseudo-residuals indicating how the current model needs to be adjusted in order to decrease the error. This fundamental concept enables the construction of weak learners and their iterative fitting to the pseudo-residuals in order to produce a strong ensemble learner for modelling complex, non-linear relationships and interactions in structured data. The model is updated additively as:

$$F_m(x) = F_{(m-1)}(x) + \gamma_m h_m(x) \quad (4)$$

3.2.4. XGBoost

XGBoost is an optimized solution that scales up gradient boosting for large data sets with high dimensionality. It grows up a tree ensemble by adding decision trees one by one, but different from many existing gradient boosting methods, XGBoost has a number of strengths: it uses a custom-designed and regularized objective function that incorporates regularization terms, it handles sparse data efficiently, and it incorporates many other “tricks” such as parallel tree construction and cache optimization. One of the key insights that enables XGBoost’s high performance is the approximation of the loss function using a second-order Taylor expansion around the current model parameters. This not only takes into account the gradient information but also the Hessian information. Thus, XGBoost is effective for problems with structured or tabular data, and it tends to overfit less than other methods [40].

3.2.5. LightGBM

LightGBM can also be classified as a gradient boosting algorithm. It is engineered to enhance training efficacy and scalability in large-scale ML problems [41]. Unlike many of the other boosting methods currently available, LightGBM utilizes a histogram-based approach to perform leaf-wise tree growth. This results in significantly reduced training complexity with minimal decrease in accuracy and greatly improved running time and memory usage, which has led to a recent surge in interest in the method and its application to real-world problems such as credit risk assessment and classification decisions regarding loan applications.

3.2.6. CatBoost

The CatBoost algorithm is a popular open-source machine-learning method based on gradient boosting over decision trees [42]. It is particularly designed for heterogeneous tabular datasets and is well-suited for problems involving categorical features. By combining ordered boosting with specialized techniques for categorical-variable processing, CatBoost reduces prediction shift and the risk of target leakage while improving predictive quality [43]. Previous studies have shown that CatBoost performs competitively on mixed-type data and can achieve strong results in financial applications, including credit-risk and fintech-lending tasks [25,43–45]. For these reasons, CatBoost was included in the present study as a methodologically relevant candidate model for loan approval prediction.

3.2.7. Support Vector Machines (SVM)

Support Vector Machines (SVMs) are a set of supervised learning algorithms, used for classification and regression tasks. SVMs determine an optimal separating hyperplane and improve the ability to generalize to new and unseen data by maximizing the margin of the classes [46]. For the input space of the original data not to be linearly separable, SVM uses kernel functions (such as the radial basis function, polynomial, or sigmoid kernels) to transform the observations into a higher-dimensional feature space where a linear separating hyperplane can be constructed more effectively [46]. The margin-maximizing approach taken by kernel methods makes Support Vector Machines (SVMs) particularly useful for classification problems with complex decision boundaries as well as for high-dimensional, structured feature representations [46,47]. Support Vector Machines (SVMs) have been used as a model for the task of credit approval prediction, due to the complex relationship between an applicant's characteristics and their creditworthiness, which might require a non-linear decision boundary in this study [48].

3.2.8. Naive Bayes Algorithm

Naive Bayes is a simple probabilistic classification method, commonly used in ML, which makes use of Bayes' theorem to calculate the posterior probability of an instance belonging to a class, given the prior probability of the class and the likelihood of the features given the class. Its scientific rigor and efficiency make it a foundational tool in statistical inference and ML [49].

For a feature vector $X = (x_1, x_2, \dots, x_p)$ and class C_k , the classifier evaluates the posterior probability as

$$P(C_k | X) \propto P(C_k) \prod_{j=1}^p P(x_j | C_k), \quad (5)$$

where the product form arises from the naive assumption that the predictors are conditionally independent given the class label. This assumption greatly simplifies probability estimation and makes the algorithm computationally efficient [50]. The Naive Bayes algorithm is one that performs well and competes favorably in many classification settings and therefore serves as an effective probabilistic baseline for credit approval prediction.

3.2.9. K-Nearest Neighbor (KNN)

K-Nearest Neighbors (KNN) is a non-parametric classification method in which an unlabeled observation is assigned to the class most frequently represented among its k – *nearest* training instances in the feature space. As an instance-based as well as lazy learner, KNN does not estimate model parameters during training, but instead relies on distance-based comparisons at the classification stage. Its simplicity and ease of implementation make it a commonly used benchmark algorithm, although its predictive performance may be affected by the choice of neighborhood size and the curse of dimensionality [51]. Therefore, for the evaluation, a simple baseline credit approval classification method was implemented using the KNN algorithm.

3.2.10. Proposed Computational Pipeline

After discussing the theoretical aspects of the individual machine-learning classifiers, the experimental procedure was organized into a unified data-mining pipeline. The step-by-step execution of this pipeline—including preliminary data preprocessing, train-test partitioning, fold-specific feature selection within a nested 3-fold cross-validation framework, GWO-based hyperparameter optimization, final model selection, hold-out test evaluation, and SHAP-based explainability analysis—is formalized in Algorithm 1. It should be noted that the leakage-control mechanism in Algorithm 1 refers specifically to the model-development stage, in which feature

selection, hyperparameter tuning, and model selection were confined to the training folds, while the independent hold-out test set was not used during these procedures.

Algorithm 1. GWO-Assisted Leakage-Aware Nested Cross-Validation Pipeline for Credit Approval Prediction

Input: $D = \{(x_i, y_i)\}_{i=1}^n$, target feature count $N = 10$,
candidate model set $\mathcal{M} = \{\text{NB, KNN, DT, RF, GBM, XGBoost, LightGBM, CatBoost, SVM}\}$,
model-specific hyperparameter spaces $\mathcal{H}_m$.

Output: selected model M^* , optimal hyperparameters θ^* , final feature subset S^* ,
test predictions $\hat{y}_{\text{test}}$, performance metrics, and SHAP values Φ .

1. Preliminary Data Preparation

- 1.1 Remove non-informative identifier customer_id.
- 1.2 Encode categorical variables into numerical form.
- 1.3 Apply Min-Max normalization as a preliminary dataset-level transformation.

2. Train-Test Partitioning

- 2.1 Split D into D_{train} (80%) and D_{test} (20%).
- 2.2 Reserve D_{test} exclusively for final hold-out evaluation.

3. Nested CV Feature Selection and GWO Optimization on D_{train}

- 3.1 For each candidate model $m \in \mathcal{M}$, define hyperparameter space $\mathcal{H}_m$.
- 3.2 For each candidate solution $\theta \in \mathcal{H}_m$ generated by GWO, maximize the objective

$$J_m(\theta) = \frac{1}{K} \sum_{k=1}^K \text{Acc}_k(m, \theta, S_k), \quad K = 3.$$

- 3.3 For each fold $k = 1, 2, 3$:
 - (a) Split D_{train} into $D_{\text{tr}}^{(k)}$ (training) and $D_{\text{val}}^{(k)}$ (validation).
 - (b) Fit Random Forest feature selector only on $D_{\text{tr}}^{(k)}$.
 - (c) Select fold-specific feature subset $S_k = \text{Top}_N(\text{Importance}_{\text{RF}}(D_{\text{tr}}^{(k)}))$ with $|S_k| = N$.
 - (d) Train model $m(\theta)$ on $D_{\text{tr}}^{(k)}[S_k]$.
 - (e) Evaluate on $D_{\text{val}}^{(k)}[S_k]$ to obtain Acc_k .
- 3.4 Retain $\theta_m^* = \arg\max_{\theta \in \mathcal{H}_m} J_m(\theta)$ for each model m .

4. Final Model Selection and Retraining

- 4.1 Select $(M^*, \theta^*) = \arg\max_{m \in \mathcal{M}} J_m(\theta_m^*)$.
- 4.2 Fit Random Forest on the full D_{train} and obtain final feature subset $S^* = \text{Top}_N(\text{Importance}_{\text{RF}}(D_{\text{train}}))$ with $|S^*| = N$.
- 4.3 Retrain M^* with θ^* on $D_{\text{train}}[S^*]$ to obtain the final model.

5. Final Hold-Out Testing and Explainability

- 5.1 Predict $\hat{y}_{\text{test}} = M^*(D_{\text{test}}[S^*])$ using the independent hold-out test set.

- 5.2 Compute performance metrics:

$$\begin{aligned} \text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, & \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}}, & \text{Specificity} &= \frac{\text{TN}}{\text{TN} + \text{FP}}, \\ \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}}, & \text{F1} &= \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}, & \text{Error Rate} &= 1 - \text{Accuracy}. \end{aligned}$$

- 5.3 Compute SHAP values Φ to explain feature contributions of M^* .

Note: Leakage control applies to fold-specific feature selection, GWO-based hyperparameter tuning, nested CV model selection, and final hold-out testing. The preliminary dataset-level normalization step is acknowledged in the manuscript discussion as a potential source of limited optimistic bias. This figure is intended as a visual summary; the formal structured pseudocode is provided in the main text.

Algorithm 1: The proposed GWO-assisted nested CV computational pipeline for credit approval prediction.

Algorithm 1. GWO-Assisted Leakage-Aware Nested Cross-Validation Pipeline for Credit Approval Prediction

Input: Loan approval dataset (D), target variable (y), candidate classifiers ($M = \{\text{DT, RF, GBM, XGBoost, LightGBM, CatBoost, SVM, NB, KNN}\}$), model-specific hyperparameter search spaces (H_m), number of selected features ($N = 10$), number of cross-validation folds ($K = 3$), GWO population size ($P = 5$), and maximum number of GWO iterations ($T = 3$).

Output: Selected optimized classifier, optimal hyperparameter configuration, final feature subset, hold-out test-set performance metrics, and SHAP-based interpretability results.

Step 1: Preliminary data preprocessing. Remove the non-informative customer_id variable. Encode categorical variables into numerical form. Apply Min-Max normalization as a preliminary dataset-level transformation. Define the predictor matrix (X) and target vector (y).

Step 2: Train-test partitioning. Split the preprocessed dataset into an 80% training set and a 20% independent hold-out test set. Reserve the hold-out test set exclusively for final evaluation and do not use it during feature selection, model selection, or GWO-based hyperparameter optimization.

Step 3: Cross-validation-based feature selection and GWO optimization on the training set. For each candidate classifier, define its model-specific hyperparameter search space. Initialize the GWO population, where each wolf

represents a candidate hyperparameter configuration. For each GWO iteration and each wolf, perform 3-fold cross-validation on the training set. Within each fold, fit the Random Forest feature selector only on the training portion of that fold, compute feature importance scores, select the top 10 features, train the candidate classifier using the wolf's hyperparameter configuration, and evaluate it on the corresponding validation fold. Use the mean validation accuracy across the three folds as the fitness value of the wolf. Update the GWO population using the alpha, beta, and delta wolves until the stopping criterion is reached.

Step 4: Model selection and final retraining. Compare the candidate classifiers according to their mean 3-fold cross-validation accuracy after GWO-based optimization and select the best-performing model. Then, fit the Random Forest feature selector on the full 80% training set to obtain the final feature subset. Retrain the selected classifier on the full training set using the optimized hyperparameters and the final selected feature subset.

Step 5: Independent hold-out testing. Evaluate the final optimized model on the independent 20% hold-out test set using the same final selected feature subset. Report accuracy, recall, specificity, F1-score, and error rate.

Step 6: Explainability analysis. Apply SHAP analysis to the final optimized model to identify the most influential predictors and to examine their global and local contributions to the model's predictions.

To enhance the clarity, reproducibility, and accessibility of the proposed computational methodology, the entire workflow is meticulously delineated in Algorithm 1 as a structured pseudocode, rather than being conveyed solely through a visual image. The algorithm explicitly specifies the input dataset, candidate model set, model-specific hyperparameter spaces, output components, preliminary data preparation, train-test partitioning, fold-specific feature selection, GWO-based hyperparameter optimization, nested cross-validation-based model selection, final retraining, independent hold-out testing, and SHAP-based explainability analysis. In this way, each stage of the proposed RF-LightGBM-GWO framework is presented in a transparent and sequential manner, allowing readers to follow, reproduce, and critically evaluate the experimental pipeline.

In Algorithm 1, the candidate models are first evaluated within a leakage-aware nested cross-validation structure. For each model, the GWO procedure searches the corresponding hyperparameter space by maximizing the mean 3-fold cross-validation accuracy computed on the training data. Within each fold, Random Forest-based feature importance is estimated only from the fold-specific training partition, and the selected feature subset is then used for model training and validation. This design ensures that feature selection and hyperparameter optimization are embedded within the model-development stage rather than being performed as external one-time preprocessing operations. After the best-performing model and its optimized hyperparameters are identified, the final model is retrained on the full training set using the selected feature subset and then evaluated on the independent hold-out test set.

It should also be noted that the accompanying figure is provided only as a visual summary of the proposed pipeline and does not replace the formal algorithmic description given in the main text. The structured pseudocode is intended to serve as the primary algorithmic specification of the method, while the figure is included to improve visual readability and to help readers understand the overall flow of the procedure. This distinction is particularly important for ensuring that the proposed method remains accessible, machine-readable, and reproducible for all readers.

3.2.11. Hyperparameter Optimization and Grey Wolf Algorithm

Hyperparameter optimization has become a focal and central component of modern ML problems and tasks. There are numerous reasons behind this methodology. First of all, model performance is usually extremely sensitive to the selection of hyperparameter values. Second, a wide range of optimization strategies has been employed for this purpose, including evolutionary and swarm-based methods such as genetic algorithms and particle swarm optimization, as well as more sample-efficient approaches such as Bayesian optimization and population-based iterative artificial bee colony algorithm [52–62]. Recent reviews highlight that these methods diverge considerably in how they explore the search space, what they cost computationally, and how well they fit different families of models and parameter landscapes. It has been argued that selecting an optimum method for hyperparameter optimization is not a purely technical detail of selecting a suitable base method for hyperparameter tuning; rather, it is a separate methodological decision that greatly affects both the achieved performance of an ML model and its achieved generalization [5].

It should be underlined that the selection of GWO as the hyperparameter optimization strategy in this study was motivated by several methodological considerations. Firstly, GWO is a derivative-free and population-based swarm intelligence algorithm; therefore, it is well suited for optimizing machine-learning models whose objective functions are non-convex, discontinuous, model-dependent, and not analytically differentiable [63,64]. This characteristic is particularly important in the present study, since nine different classifiers with heterogeneous hyperparameter spaces were optimized under a unified experimental framework. Previous studies and reviews on

hyperparameter optimization have emphasized that the choice of optimization strategy can substantially affect model performance, computational cost, and generalization behavior [5,64].

Compared with Bayesian Optimization, GWO does not require the construction of a surrogate probabilistic model of the objective function. Bayesian Optimization is generally recognized as a sample-efficient strategy, particularly in relatively low-dimensional and expensive black-box optimization problems; however, its performance may depend on the choice of surrogate model, acquisition function, and the structure of the search space [65–67]. In the present multi-model setting, where tree-based ensembles, kernel-based methods, instance-based learners, and probabilistic classifiers were optimized under heterogeneous and mixed-type hyperparameter spaces, GWO provided a simpler and more uniformly applicable optimization mechanism. Compared with Particle Swarm Optimization (PSO) and Genetic Algorithms (GA), GWO has a relatively simple control structure and fewer algorithm-specific parameters to tune [50,63,68]. Its alpha–beta–delta leadership hierarchy enables the search process to balance exploration and exploitation by allowing the three best candidate solutions to guide the remaining search agents [51]. Thus, this mechanism reduces dependence on a single best solution and supports stable search behavior during hyperparameter tuning [63,64]. For these reasons, GWO was selected as a practical, flexible, and computationally manageable optimization method for the proposed multi-model credit approval prediction framework.

3.2.12. Grey Wolf Optimizer (GWO)

GWO is a swarm-based, derivative-free method developed by Mirjalili et al. (2014). The inspiration for this method comes from the leadership hierarchy and hunting mechanism of grey wolves [63]. The solutions are divided into four categories (alpha, beta, delta, omega), and the search is based on the wolf's behavior. The hunting process can be understood as a natural sequence consisting of three interconnected phases: initially, tracking and approaching the target prey, then encircling it to limit its movement, and finally, executing the attack. It is highly important to state that GWO belongs to the family of swarm intelligence algorithms, similar to Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Artificial Bee Colony Algorithm (ABC) [69].

In this research, GWO was employed to fine-tune the hyperparameters of all potential machine-learning algorithms. Each wolf symbolized a potential combination of hyperparameters tailored to a specific model, with an independent GWO optimization process carried out for each classifier based on its designated hyperparameter space. The fitness of each candidate solution was identified according to the validation performance of its corresponding model, evaluated under the selected criterion. During the search process, the three leading wolves guided the position updates of the remaining population, allowing the optimization procedure to balance global exploration with local exploitation [63,64].

In the original GWO framework, this search balance is controlled by a coefficient parameter that decreases over the course of iterations, thereby encouraging broader exploration in the early stages and more intensive exploitation in later stages. The optimization process was terminated when the predefined stopping criterion was reached, and the best-performing wolf was retained as the final hyperparameter configuration for the corresponding model. Many studies have applied the Grey Wolf Optimizer (GWO) technique to various ML tasks, including hyperparameter tuning, demonstrating improvements for performance metrics [60,63,64].

4. FINDINGS AND RESULTS

This section presents the results of the empirical-based structure for credit forecasting tasks. Firstly, we present the information about the data used in the research. Secondly, we explain how we selected features to use as input for the ML models. The dynamic feature selection process is integrated within a nested cross-validation framework to prevent data leakage. In addition, the comparative performance of nine distinct ML algorithms optimized via GWO is applied. Finally, we shortly report on the results of the performance metrics of the models, and also present an Explainable AI (XAI) through SHAP (SHapley Additive exPlanations) visualization to illustrate the most important input features for the selected models.

4.1. Data Acquisition & Dataset Characteristics and Data Preprocessing

The first part of this study is an acquisition of data. It has been used the Realistic Loan Approval Dataset | US & Canada, downloaded from Kaggle, a website dedicated to hosting datasets for data science studies. The Realistic Loan Approval Dataset can be downloaded from here: https://www.kaggle.com/datasets/parthpatel2130/realistic-loan-approval-dataset-us-and-canada (Patel, 2025). Before model development, the customer_id variable was removed because it is a unique identifier and does not carry predictive information. Its inclusion could introduce artificial bias into the learning process. The dataset was

examined for missing values, and no missing observations were detected. Outlier analysis was then conducted using the Interquartile Range (IQR) method. Although some extreme values were observed in continuous financial variables such as age, annual income, credit score, and current debt, these observations were retained because they may represent meaningful rare financial profiles rather than erroneous records. It must be noted that the dataset contains 50,000 samples and 18 features after removing the non-informative customer_id attribute.

Categorical variables were transformed into numerical representations, and numerical variables were scaled into the [0, 1] interval using Min-Max normalization. Following preprocessing, the dataset was split into an 80% training set and a 20% independent hold-out test set. All model selection, dynamic feature selection, and GWO-based hyperparameter optimization procedures were performed only within the training set, while the hold-out test set was kept completely unseen until the final evaluation stage.

The second step of our experimental methodology is the data preprocessing step. First of all, we remove the customer_id feature, containing all unique values, in order to break the bias in the used algorithms. The feature completeness is high, i.e., there are absolutely no missing values in our dataset. It has been performed an exploratory data analysis to verify this. After that, to identify potential outliers in the dataset, an outlier analysis based on the Interquartile Range (IQR) was performed for the categorical and structural features. Outliers were not found for most of the features, but extreme values were present in the continuous financial features (age, annual_income, credit_score, current_debt). Even though the outliers were kept in the dataset, the majority of the ML algorithms used in this study, specifically tree-based and gradient boosting frameworks, are robust to outliers. By removing the outliers, valuable financial behaviors and rare risk profiles could be lost. Thus, they were intentionally retained in the dataset. To finalize the data preparation, categorical variables were encoded into machine-readable formats. These procedures include transforming all numerical variables into the [0, 1] range using Min-Max Normalization. This enforces scale uniformity on the underlying feature data. Consequently, integer features with larger numeric ranges cannot dominate the model. Following these preprocessing steps, to minimize the risk of information leakage during the model-development stage, the dataset was partitioned into an 80% training set and a 20% hold-out test set. All subsequent dynamic feature selection and hyperparameter optimization procedures were strictly confined to this training set, ensuring the test data remained completely unseen.

4.2. Data Visualization, Feature Selection

To identify the relationships between the variables in the financial data, a correlation matrix was produced and illustrated (See Figure 2). The correlation matrix revealed the linear relationships between several variables, such as borrower characteristics and loan approval status, indicating the potential for multicollinearity.

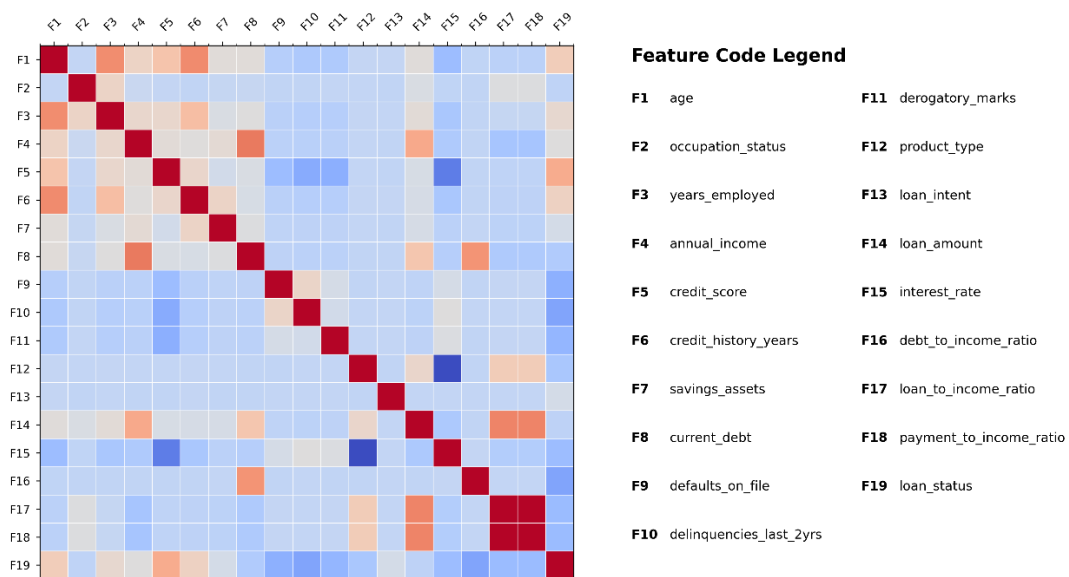


Figure 2: Correlation matrix of the dataset features.

Following the initial visualization, a rigorous feature selection process was executed to determine the most influential variables while strictly preventing data leakage.

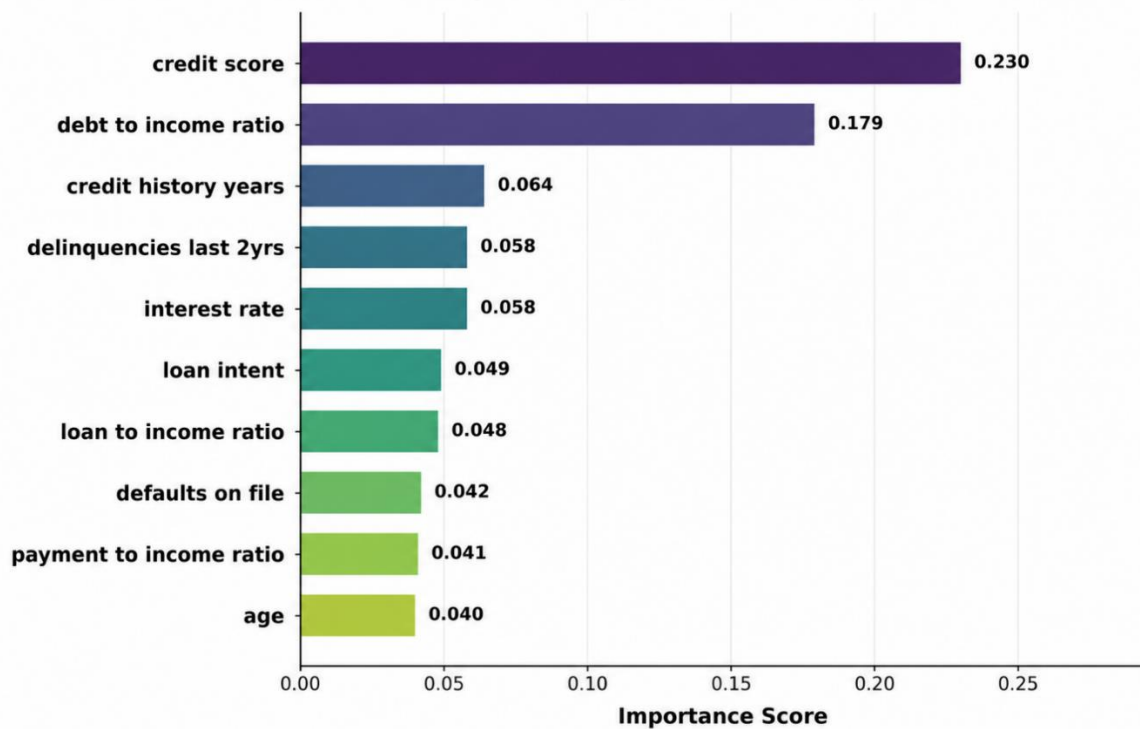


Figure 3: Top 10 feature importances determined by the Random Forest algorithm.

Random Forest algorithm was employed as a supervised feature-selection tool in order to identify the most informative predictors for loan approval classification. In this study, feature selection was not performed once on the full dataset or on the entire 80% training partition as a fixed pre-processing step. Instead, to reduce the risk of information leakage and selection bias, the feature-selection procedure was embedded within the model-development stage and applied exclusively within the training data. More specifically, during the nested cross-validation process, Random Forest was fitted separately for each training fold, and feature importance scores were computed to rank the predictors by their contribution to classification performance. Based on these fold-specific importance scores, the top 10 features were selected dynamically and passed to the subsequent hyperparameter optimization and classification stages. In this way, feature selection was treated as an internal component of the training pipeline rather than as an external dimensionality-reduction step. This design ensured that no information from the validation or hold-out test data was used during feature selection. Feature selection was performed on the data without reference to the validation or hold-out test data. It therefore gives an indication of the relative importance of the different features without the benefit of any information on the model's future performance. Based on this selection process, the four most important predictors of loan approval (in terms of likelihood of the loan being granted) across all used folds turned out to be: `credit_score`, `debt_to_income_ratio`, `credit_history_years`, and `defaults_on_file`. Again, these variables are the key predictors of loan approval.

4.3. Comparative Performance of Classification Algorithms

In this stage of the research, a detailed comparative analysis was conducted. Nine different classification algorithms—ranging from traditional statistical models to advanced gradient boosting frameworks—were evaluated. Unlike standard approaches that rely on default parameters, this study subjected all candidate models to Grey Wolf Optimization (GWO) within a rigorous nested 3-Fold Cross-Validation (CV) framework. During this nested process, both the extraction of the top 10 features and the hyperparameter tuning were dynamically executed exclusively within the training folds. This leakage-aware methodology was designed to minimize information leakage during feature selection, model comparison, hyperparameter tuning, and final hold-out evaluation, while improving robustness against overfitting.

It must be acknowledged that 3-fold nested cross-validation was adopted as a practical balance between validation reliability and computational feasibility. In the proposed framework, cross-validation was not used as a simple model evaluation procedure only. Instead, it was embedded within a more computationally demanding pipeline that included fold-specific dynamic feature selection, GWO-based hyperparameter optimization, and the

comparative evaluation of nine different classification algorithms. Since each candidate GWO solution required repeated model training and validation inside the cross-validation loop, increasing the number of folds to 5 or 10 would have substantially increased the computational burden.

Moreover, the main objective of the cross-validation stage was model selection and hyperparameter optimization within the training data, while the final generalization performance was independently assessed using a completely unseen 20% hold-out test set. Therefore, the 3-fold nested cross-validation procedure was used to reduce selection bias and guide the optimization process, whereas the final reported performance metrics were obtained from the independent test set. This design ensured that the validation process remained computationally manageable while preserving a strict separation between model development and final testing.

For reproducibility, a separate GWO search was performed for each classifier using a predefined algorithm-specific hyperparameter space. The population size was set to 5 wolves, and the maximum number of iterations was fixed at 3. We want to highlight that these values were selected to maintain computational feasibility within the multi-model nested optimization setting. To minimize data leakage, the fitness value was defined as the mean 3-fold nested cross-validation accuracy computed on the training set, with the top 10 features dynamically selected within each fold. A fixed random seed (42) was used across optimization runs to ensure consistency. The search ranges included, depending on the classifier, parameters such as tree depth, number of estimators, learning rate, cost parameter, kernel parameter, and neighborhood size.

Thus, the true generalization capacity of these GWO-optimized algorithms was systematically measured. As demonstrated in Table 2, modern ensemble techniques significantly outperformed conventional algorithms. Specifically, the GWO-optimized LightGBM emerged as the top-performing model in this rigorous evaluation, achieving a peak 3-Fold CV accuracy of 91.13%. It was closely followed by CatBoost (91.00%) and XGBoost (90.84%). Conversely, simpler baseline models like Naive Bayes struggled to capture the complex, non-linear financial patterns, yielding a substantially lower accuracy of 68.90%. The results clearly illustrate that optimized gradient boosting ensembles maintain a highly robust and superior performance for credit risk assessment. The optimized hyperparameter values obtained for each candidate model are reported in Table 2.

Table 2: 3-Fold Nested Cross-Validation accuracies and GWO-optimized hyperparameters of the candidate algorithms

<i>Model</i>	<i>3-Fold CV Accuracy (%)</i>	<i>Optimized Hyperparameters (via GWO)</i>
<i>LightGBM</i> (best cross-validation model)	91.13	n_estimators=242, learning_rate=0.0791, max_depth=4
<i>CatBoost</i>	91.00	iterations=191, learning_rate=0.1077, depth=6
<i>XGBoost</i>	90.84	n_estimators=200, learning_rate=0.1182, max_depth=6
<i>GBM</i>	90.82	n_estimators=178, learning_rate=0.0750, max_depth=6
<i>Random Forest</i>	89.81	n_estimators=288, max_depth=13
<i>SVM</i>	88.23	C=2.85, gamma=0.1963
<i>Decision Tree</i>	87.75	max_depth=11
<i>KNN</i>	86.97	n_neighbors=12
<i>Naive Bayes</i>	68.90	Default Parameters

It is vital to underline that the inclusion of Table 2 is considered methodologically essential, as the entire model selection process is documented within it in a transparent and reproducible manner. Instead of being selected arbitrarily, the final classifier was determined through a systematic and well-defined evaluation procedure. Within this procedure, nine candidate algorithms were initially assessed under a unified GWO-based hyperparameter optimization strategy. In parallel, a nested 3-fold cross-validation setting was employed, by which the risk of overfitting and biased model selection was effectively minimized. In Table 2, both the validation accuracies and the optimized hyperparameter configurations are reported for each of the candidate models. By this means, the relative performance of the algorithms can be examined by the reader under fully comparable experimental conditions. As clearly indicated by the reported results, the highest cross-validation accuracy was achieved by LightGBM. On the basis of this empirical finding, the selection of LightGBM as the core classifier of the proposed

RF-LightGBM-GWO hybrid framework is firmly justified. Accordingly, Table 2 should not be regarded as a merely descriptive component of the study. Instead, it should be considered that the evidentiary foundation upon which the transition from multi-model benchmarking to the final model evaluation, subsequently presented in Table 3, is deliberately and meticulously grounded.

4.4. RF-LightGBM-GWO Model and Artificial Intelligence Explainability (XAI)

The RF-LightGBM-GWO hybridization successfully optimized the model, and its true generalization capability was evaluated on the unseen 20% test set. The final model demonstrated highly robust predictive power, achieving an Accuracy of 92.17%, a Recall of 93.46%, an F1-Score of 92.91%, a Specificity of 90.60%, and an Error Rate of 7.83% (See Table 3).

Table 3 : RF-LightGBM-GWO Hybrid Model Performance Evaluation Metrics

<i>RF-LightGBM-GWO Hybrid Model</i>	<i>Performance Evaluation Metrics</i>
<i>Accuracy</i>	92.17 %
<i>Recall (Sensitivity)</i>	93.46 %
<i>Specificity</i>	90.60 %
<i>F1-Score</i>	92.91 %
<i>Error Rate</i>	7.83 %

While complex ensemble models are known to deliver superior predictive performance, their lack of transparency—often referred to as the "black-box" problem—poses a substantial challenge in terms of regulatory compliance for financial institutions. To mitigate this issue, SHAP was incorporated into the proposed framework [70,71]. As depicted in the SHAP summary plot (Figure 4), variables such as credit_score and debt_to_income_ratio have the highest impact on the LightGBM model's output.

The SHAP analysis not only corroborates the precise Random Forest–based feature selection performed within the nested training folds but also offers fine-grained transparency, ensuring that the credit approval decisions made by the hybrid model are interpretable. It can also be interpreted as financially well-grounded.

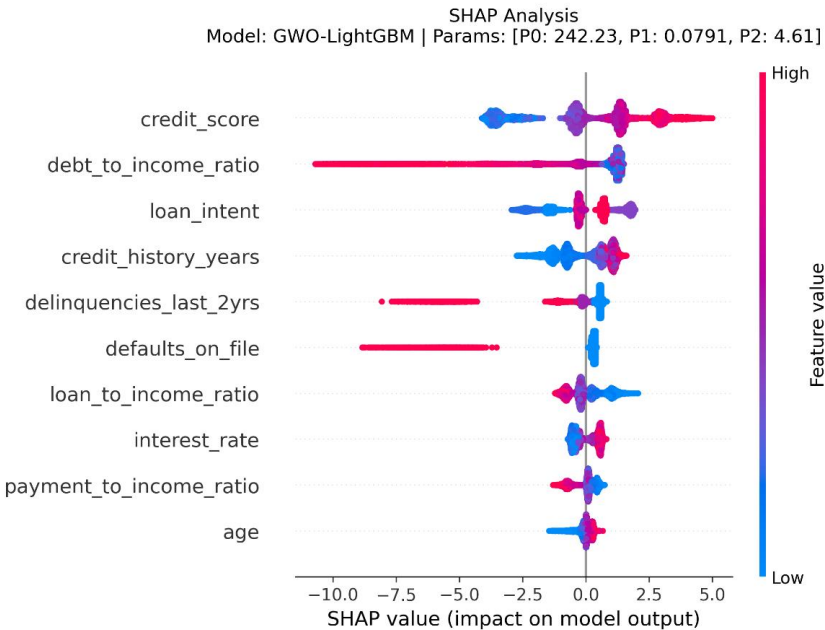


Figure 4: SHAP summary plot demonstrating global feature importance and directional impact for the LightGBM model.

5. Discussion

This study developed an explainable hybrid machine-learning framework for loan approval prediction by combining comparative model benchmarking, Grey Wolf Optimizer (GWO)-based hyperparameter tuning, dynamic feature selection, and SHAP-based interpretability analysis. The experimental findings demonstrated that ensemble-based learning methods generally outperformed conventional classifiers in the credit approval prediction task. Among the candidate algorithms, the RF-LightGBM-GWO hybrid framework provided strong and balanced predictive performance on the independent unseen test set. In addition, SHAP analysis showed that variables such as credit score and debt-to-income ratio played a dominant role in the final model decisions, supporting the financial interpretability of the proposed framework.

The findings of this study demonstrate that sophisticated machine-learning techniques can yield robust performance in predicting loan approvals. In general, ensemble-based models performed better than simpler and more traditional classifiers. Among the algorithms that were evaluated, the strongest performance was delivered by CatBoost, XGBoost, GBM, and LightGBM. In contrast, a noticeably lower level of performance was observed for Naive Bayes and K-Nearest Neighbors.

Building on the performance differences reported above, it was clearly shown by these results that modern credit data are more effectively modeled by algorithms in which nonlinear relationships can be captured. Furthermore, the complex interactions among borrower characteristics were found to be more comprehensively represented by these models, thereby reinforcing their suitability for contemporary credit risk assessment. We want to emphasize that one of the most important findings of the study was the performance of the RF-GWO-optimized LightGBM model, which achieved the highest 3-fold cross-validation accuracy (91.13%) among the candidate algorithms. Strong predictive performance was likewise produced by the final RF-LightGBM-GWO hybrid model on the previously unseen 20% test set. An accuracy of 92.17% was achieved by the proposed framework on this independent subset. In addition, a recall of 93.46% was achieved, clearly reflecting the model's ability to correctly identify approved applicants. A specificity of 90.60% was also observed, indicating that rejected cases were reliably distinguished as well. Furthermore, an F1-score of 92.91% was obtained, through which the harmonic balance between precision and recall was confirmed. These results effectively demonstrated a well-balanced capability to correctly classify both approved and rejected loan applications. Such a balance is considered to be of particular importance within the domain of lending decisions.

More specifically and precisely, this is mainly because reliable applicants are expected to be approved with high accuracy, while risky applicants must simultaneously be detected with sufficient sensitivity. From this perspective, a practical and robust solution for credit approval prediction appears to be provided by the proposed hybrid framework. Another important contribution of the study is the inclusion of explainable AI. In financial applications, strong predictive performance alone is often not enough; decision makers also need to understand why a model produces a certain outcome. This issue was effectively addressed through the application of SHAP analysis. By means of this analysis, credit score and debt-to-income ratio were identified as the most influential variables in the final predictions. The financial relevance of these findings is widely acknowledged. In addition, the overall transparency of the model is meaningfully enhanced through such interpretability insights. As a result, it is clearly demonstrated by this study that predictive strength and interpretability can be jointly considered. Rather than being treated as separate or mutually exclusive objectives, these two aspects are shown to be effectively reconciled within a single, well-structured analytical framework.

The findings indicate that combining ensemble learning, metaheuristic optimization, and explainable AI can provide a practical and transparent decision-support framework for financial institutions. Unlike a purely black-box predictive model, the proposed pipeline not only improves classification performance but also provides interpretable explanations regarding the factors that influence approval and rejection decisions. This is particularly important in credit risk assessment, where predictive accuracy, transparency, and regulatory accountability should be considered together.

Future research can extend the present study in several directions. First, the proposed framework should be validated on larger and institution-level real-world credit datasets collected from different financial markets to examine its generalizability across various borrower populations and lending environments. Second, future studies may compare GWO with other optimization strategies such as Bayesian Optimization, Particle Swarm Optimization, Genetic Algorithms, Artificial Bee Colony, and hybrid metaheuristic approaches under the same experimental conditions. Such comparisons would provide a deeper understanding of the relative strengths and limitations of each optimization strategy.

Third, future work should investigate model fairness and bias more comprehensively by examining whether the prediction outcomes differ across demographic or socio-economic subgroups. Fairness-aware learning

strategies may be integrated into the model development process to reduce potential discriminatory effects. Fourth, calibration analysis should be performed to assess whether the predicted probabilities accurately reflect real-world approval likelihoods, especially when the model is intended to support risk-sensitive financial decisions. Finally, future studies may incorporate additional explainability tools, local explanation methods, and regulatory-oriented reporting mechanisms to improve the practical adoption of the proposed framework in real financial decision-making systems.

Despite these strengths, some limitations should be acknowledged. First, the dataset was obtained from Kaggle rather than from an active financial institution, which may limit the direct generalizability of the results. Second, although GWO was successfully used for hyperparameter tuning, alternative optimization strategies may lead to different results. Third, it should be clarified that the leakage-prevention pipeline described in Algorithm 1 mainly refers to the model-development stage, including fold-specific feature selection, GWO-based hyperparameter optimization, nested cross-validation-based model selection, and final evaluation on the independent hold-out test set. In these stages, no validation or hold-out test information was used during feature selection, hyperparameter tuning, or model selection. However, an initial dataset-level preprocessing stage was applied before model development, including the removal of non-informative variables, categorical encoding, and Min-Max normalization. Although this preprocessing strategy provided a consistent feature representation across all candidate models, the dataset-wide normalization step may have introduced a limited degree of optimistic bias. Therefore, future work should implement all learned preprocessing transformations, including normalization parameters, strictly within the training folds and then apply the fitted transformations to the validation and test sets [54,57,58,62].

5. Conclusion

This study developed an explainable hybrid ML framework for loan approval prediction by combining comparative model benchmarking, Grey Wolf Optimizer-based hyperparameter tuning, and SHAP-based interpretability analysis. The findings showed that ensemble-based methods consistently outperformed conventional classifiers, with the final RF-LightGBM-GWO hybrid model delivering high and balanced predictive performance on the unseen test set. In addition, SHAP analysis revealed that credit score and debt-to-income ratio were the most influential predictors, providing transparent and financially meaningful insight into the model's decisions. Overall, the results suggest that integrating advanced ensemble learning, systematic hyperparameter optimization, and explainable AI can offer a robust and practical framework for improving credit approval decision-making. On the other hand, future studies should validate this approach on larger real-world datasets and examine fairness, calibration, and regulatory transparency in greater depth.

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